

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.6	8.7	1.5	0.1	0.2	56.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.7	7	0.7	0.2	0	56.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.8	8.9	1.2	0.3	0.3	56.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.3	6.8	0.3	0.1	0	56.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.6	9	1.3	0.3	0.1	56.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.5	7.5	0.3	0.2	0	56.5
U2 - Efficacy additive alpha prior Beta(1, 1)	31.8	9.3	1.3	0.2	0.2	57.2
U3 - Efficacy additive alpha prior Beta(1, 1)	35.1	7.8	0.6	0.3	0	56.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	4.2	2.2	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	4.3	2.2	0.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15	4.3	2.3	0.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15	4.3	2.3	0.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.1	4.3	2.2	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15	4.3	2.2	0.8	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.1	4.2	2.2	0.8	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	15	4.2	2.3	0.9	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2	0.5	0.3	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	36.3	17	31.3	10.5	3	1.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.9	18.9	28.7	9.6	2	1.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.8	17.9	30.9	11	2.4	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	38.6	18.5	29.4	9.4	2.1	2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.1	17.1	31.3	10.9	2.7	1.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.5	19.3	29.1	9	2.2	1.9
U2 - Efficacy additive alpha prior Beta(1, 1)	35.4	17.6	32.5	10.2	2.4	1.9
U3 - Efficacy additive alpha prior Beta(1, 1)	38.7	19.4	29.3	8.4	2.2	2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.5	5.7	6.8	3.7	2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.6	5.7	6.7	3.8	2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.5	5.7	6.8	3.9	1.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	5.7	6.8	3.8	2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	5.8	6.7	3.8	2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	5.8	6.7	3.7	2	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.5	5.7	6.7	3.9	1.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.5	5.7	6.8	3.7	1.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.5	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.5	1.3	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.4	1.3	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.5	1.3	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.4	1.2	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.5	1.2	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.5	1.3	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.5	1.2	0.8	0.3	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19	39.5	21.6	8.4	1.5	10
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.3	42.7	18.7	7	1.1	10.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	38.9	21.9	7.9	1.6	10.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.8	41.4	19.6	7.2	1	10
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.8	38.8	23.4	7.5	1.5	10
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.9	42	19.5	6.4	1	10.2
U2 - Efficacy additive alpha prior Beta(1, 1)	18.5	39.8	22.8	7.4	1.4	10.1
U3 - Efficacy additive alpha prior Beta(1, 1)	20.3	43.2	19.4	6.3	0.9	9.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	7.7	6.1	3.6	1.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	7.8	6	3.6	1.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	7.7	6.1	3.6	1.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	7.6	6.2	3.6	1.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	7.6	6.2	3.6	1.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	7.5	6.2	3.6	1.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	7.7	6.2	3.5	1.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.6	7.8	6	3.6	1.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.1	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.2	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	1.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	1.2	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3	1.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	1.2	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.2	1	0.3	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	8.4	23.6	41.1	16.1	1.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.1	10.8	19.9	39.8	13.5	1.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	9.3	22.9	41.1	15.4	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	9.8	19.5	40.6	13	1.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	8.8	22.7	40.4	16.5	1.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	10.8	19.5	39.6	12.4	1.8
U2 - Efficacy additive alpha prior Beta(1, 1)	9.1	9.4	21.9	42.2	15.6	1.8
U3 - Efficacy additive alpha prior Beta(1, 1)	15.4	10.6	19.5	40.1	12.6	1.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	4.5	6.4	8.2	6.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	4.5	6.4	8.1	6.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	4.4	6.5	8.2	6.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	4.4	6.4	8.2	6.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	4.4	6.4	8.2	6.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	4.5	6.5	8.2	6.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.5	4.4	6.5	8.2	6.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.5	4.4	6.4	8.3	6.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.4	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.4	2.3	0.8	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	31.1	29.2	20.3	10.1	4.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	36.8	25.1	18.8	5.6	4.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	31.9	27.6	20.8	10.5	4.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	35.1	26.3	19.2	5.8	4.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	31.2	28	21.4	10.3	4.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	36.6	25.9	18.1	5.7	4.2
U2 - Efficacy additive alpha prior Beta(1, 1)	5.2	32.5	27.4	21	9.7	4.2
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	36.3	25.8	19.2	5.6	4.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	6.3	6.7	6	5.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	6.3	6.6	6	5.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	6.2	6.7	6	5.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	6.2	6.7	6	5.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	6.3	6.6	6	5.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	6.2	6.7	6	5.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.9	6.3	6.6	6	5.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.9	6.2	6.6	6	5.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	1.7	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.3	1.7	2.2	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	1.7	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	1.7	2.2	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.2	1.7	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.7	2.2	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	1.7	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	1.7	2.2	1.1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	68.3	13.3	2.6	1.4	0.4	14
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	71.2	11.5	1.9	0.9	0.4	14.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.5	13.1	2.9	1	0.6	13.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	69.5	12.6	2.4	1.2	0.4	13.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.6	13.2	3.2	1.1	0.4	13.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	71	11.2	2.6	0.9	0.4	13.9
U2 - Efficacy additive alpha prior Beta(1, 1)	69.7	11.5	2.9	1.5	0.4	14
U3 - Efficacy additive alpha prior Beta(1, 1)	70.1	12.5	2.1	1.1	0.3	13.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.3	4.3	2.2	1.1	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.2	4.4	2.2	1	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.4	4.3	2.2	1.1	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.3	4.4	2.2	1.1	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.4	4.3	2.2	1.1	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.3	4.4	2.2	1	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.2	4.5	2.2	1.1	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.3	4.4	2.3	1.1	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.6	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.6	0.4	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.6	0.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.4	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	1.6	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.4	0.3	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.5	36.2	21.8	9.6	3.4	2.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.6	39.5	19.5	8	1.9	2.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.7	36.6	20.8	10.2	3.1	2.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.6	39.2	19.1	8.9	1.7	2.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.4	34.9	21.6	10.5	3.1	2.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.8	39	18.5	9	2.2	2.5
U2 - Efficacy additive alpha prior Beta(1, 1)	27.1	35.9	20.8	10.3	3.4	2.5
U3 - Efficacy additive alpha prior Beta(1, 1)	27.9	40.2	18.5	9.1	1.9	2.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	7.3	5.6	3.5	2.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	7.2	5.7	3.6	2.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	7.1	5.6	3.7	2.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	7.3	5.6	3.6	2.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	7.1	5.8	3.7	2.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	7.2	5.6	3.7	2.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.8	7.1	5.6	3.7	2.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.7	7.3	5.6	3.7	2.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.1	0.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1	1	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1	0.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1.1	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1	1	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1	1	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1	1	0.4	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	44	8	2.9	1.3	0.3	43.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	46.3	6.9	2.4	1.1	0.2	43.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	44.6	8.4	2.6	0.8	0.5	43.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	46	8.1	1.8	0.8	0.2	43.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	43.3	8.7	3.1	1.1	0.4	43.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	45.6	8.3	1.4	1.1	0.2	43.4
U2 - Efficacy additive alpha prior Beta(1, 1)	43.9	8.8	2.6	1.1	0.3	43.3
U3 - Efficacy additive alpha prior Beta(1, 1)	45.8	8.3	1.6	0.8	0.2	43.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17	4	2	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	3.8	2.1	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.1	3.9	1.9	0.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.3	3.7	2	0.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17	3.9	2	0.7	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.1	4	1.9	0.7	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.2	3.9	2	0.7	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.1	4	2	0.7	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.4	0.2	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.4	11.5	15	12.4	23	18.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.3	12.8	13.9	9.8	20.6	18.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	12.6	16.2	10.6	23.4	18.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.2	12.8	13.3	10.9	20.2	18.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.8	12.8	16.2	11.2	23.6	18.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23	13.8	12.9	10.5	21	18.8
U2 - Efficacy additive alpha prior Beta(1, 1)	18.9	12	16.4	11.6	22.8	18.3
U3 - Efficacy additive alpha prior Beta(1, 1)	24	12	14.4	10.7	20.5	18.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	4.7	5.2	3.9	5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	4.8	5.2	3.9	5.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	4.7	5.3	3.9	5.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	4.6	5.2	3.9	5.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	4.7	5.4	3.8	5.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	4.7	5.3	3.9	5.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	4.7	5.3	3.8	5.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	4.7	5.4	3.8	5.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.7	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.7	1.3	1.4	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.7	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.7	1.3	1.4	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.7	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.7	1.3	1.4	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.7	1.3	1.4	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.6	1.4	1.4	0.6	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.8	29.4	11.8	4.8	1.8	37.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.3	31.7	10.9	4.4	1.5	37.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14	30.5	11.6	4.8	1.6	37.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.2	32.4	10	4.6	1.4	37.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.1	30.1	12.6	4.2	1.8	37.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	31.6	10.8	4.2	1.5	37.3
U2 - Efficacy additive alpha prior Beta(1, 1)	14.4	29.6	11.9	4.7	2	37.4
U3 - Efficacy additive alpha prior Beta(1, 1)	14.5	30.6	11.5	4.3	1.4	37.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.6	6.4	4.1	2.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.5	6.6	4	2.1	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	6.3	4.1	2.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	6.6	3.9	2.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	6.3	4.2	2.1	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	6.4	4.1	2.2	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.6	6.4	4	2.1	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.6	6.5	4.1	2.1	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.3	0.8	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.7	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.2	0.8	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.7	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.7	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.7	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.3	0.7	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.3	0.7	0.6	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.2	19.1	4.4	0.5	0	6.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	71.1	19.3	2.6	0.4	0	6.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	69	20.5	3.3	0.4	0	6.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	71.8	18.7	2.4	0.4	0	6.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	69	20.4	3.8	0.3	0	6.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	72.3	18.7	2	0.3	0	6.7
U2 - Efficacy additive alpha prior Beta(1, 1)	69.8	19.8	3	0.5	0	6.9
U3 - Efficacy additive alpha prior Beta(1, 1)	71.6	18.7	2.7	0.4	0	6.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	6.1	3.2	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.2	6.2	3.3	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.2	6.3	3.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.2	6.2	3.4	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	6.4	3.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	6.3	3.4	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.2	6.3	3.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.3	6.2	3.3	1	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.4	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.4	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.4	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.4	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.4	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.4	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.6	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.4	0.7	0.3	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.6	14.8	11.3	34.7	19.4	3.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.5	15.9	7.6	31.3	17.4	3.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17	14.8	10.6	35.9	18.4	3.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.2	15.9	7.7	31.9	17.1	3.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18	13.8	11.5	34.3	19.1	3.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.6	15.5	8.1	33.1	16.4	3.3
U2 - Efficacy additive alpha prior Beta(1, 1)	16.8	15.2	10.7	34.4	19.7	3.2
U3 - Efficacy additive alpha prior Beta(1, 1)	24.6	14.9	7.7	32.4	17.1	3.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	5	5.3	7.4	6.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	4.9	5.4	7.3	6.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	5	5.3	7.4	6.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	4.9	5.4	7.4	6.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	4.9	5.4	7.4	6.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	4.9	5.4	7.3	6.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.1	5	5.4	7.3	6.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.1	4.9	5.3	7.4	6.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1.4	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3	1.4	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3	1.4	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3	1.4	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3	1.4	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3	1.4	2.3	0.8	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.5	29.8	14.1	4.9	2	32.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	32.4	11	5.1	1	33
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.4	30.4	13.6	5	1.8	32.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.8	30.9	12.2	4	1.4	32.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.1	30.9	13.4	5	1.8	32.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	33	10.8	4.5	1.6	32.7
U2 - Efficacy additive alpha prior Beta(1, 1)	16.1	30.9	13.2	4.8	1.6	33.4
U3 - Efficacy additive alpha prior Beta(1, 1)	18	32.3	11.9	4.3	1.3	32.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	6.5	4.5	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	6.5	4.4	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	6.5	4.5	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	6.5	4.5	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	6.6	4.5	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	6.5	4.4	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.3	6.6	4.3	2.4	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.2	6.7	4.3	2.5	1.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.6	0.9	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	0.9	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.9	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.6	0.9	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.6	0.9	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.6	0.9	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.9	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.6	0.9	0.7	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.3	16.5	15.2	22.8	13.7	7.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32	16.4	11.5	21.5	11.1	7.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.7	15.7	16	22.2	13.8	7.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.3	17.4	11	20.9	10.9	7.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.6	15.8	16.7	22.8	14.6	7.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.9	16.7	10.4	21.6	10.9	7.5
U2 - Efficacy additive alpha prior Beta(1, 1)	22.3	17.3	16.2	22.8	14	7.4
U3 - Efficacy additive alpha prior Beta(1, 1)	33.1	16.7	11.3	20.6	10.9	7.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7	5.6	5.8	5.8	5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7	5.7	5.6	5.8	5.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7	5.5	5.7	5.8	5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7	5.6	5.7	5.8	5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7	5.5	5.7	5.9	5.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7	5.5	5.7	5.9	5	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.1	5.5	5.7	5.7	5	
U3 - Efficacy additive alpha prior Beta(1, 1)	7	5.5	5.7	5.9	5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.2	1.5	1.9	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.2	1.5	2	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.2	1.5	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.2	1.5	2	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.2	1.5	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.2	1.5	2	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.2	1.5	1.9	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.2	1.4	2	0.7	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	51.8	28.5	8.8	2.7	1.1	7.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	60.9	23.6	6.4	1.1	0.7	7.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	50.6	28.7	9.4	3.1	0.9	7.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	61.8	24.6	5.1	1.1	0.4	7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	50.8	28.8	8.8	3.3	1.1	7.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	61.6	23.9	5.5	1.2	0.5	7.3
U2 - Efficacy additive alpha prior Beta(1, 1)	51	28.4	8.9	3.5	1	7.2
U3 - Efficacy additive alpha prior Beta(1, 1)	60.9	25	5.2	1.1	0.7	7.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.3	7.4	5.6	3.2	1.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.3	7.4	5.6	3.2	1.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.3	7.4	5.6	3.2	1.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.3	7.4	5.6	3.1	1.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	7.3	5.6	3.2	1.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	7.3	5.6	3.2	1.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.4	7.5	5.5	3.1	1.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.3	7.5	5.5	3.2	1.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.5	1.5	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.5	1.5	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.5	1.4	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	3.5	1.5	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.5	1.5	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.4	1.5	1.3	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.5	1.4	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.5	1.4	1.3	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.6	26.3	17.8	16.7	10.5	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.5	29.7	16.1	13.1	5.5	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.3	26.1	18	16.7	10.8	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.6	30.5	16.6	11.7	5.5	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.5	26.6	17.9	16.9	11	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.8	30.5	15.7	13.9	5	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	28.1	25	19.2	16.4	11.2	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	36.3	29.6	15.6	13.3	5.1	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.9	5.7	5.5	5.8	7.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6	5.7	5.3	5.7	7.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	5.7	5.4	5.8	7.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6	5.7	5.3	5.8	7.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	5.6	5.3	5.8	7.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	5.7	5.4	5.8	7.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	6	5.7	5.4	5.7	7.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	6	5.7	5.4	5.8	7.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.2	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.2	1.8	2.6	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.2	1.8	2.7	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.3	1.8	2.6	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.2	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.2	1.8	2.6	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.2	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.2	1.8	2.7	1.5	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	32	50.3	16	1.4	0	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.3	49	14	1.4	0	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.6	50.9	15.6	1.6	0	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.4	50.3	13.7	1.3	0	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.8	50.4	15	1.5	0	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.1	49.7	13.9	1	0	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	32.8	49.5	16.1	1.3	0	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	34.7	50.1	13.6	1.3	0	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	10.2	6.9	2.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10	10.2	7.1	2.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	10.1	7	2.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	10.2	6.9	2.3	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	10.2	7	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	10.3	7	2.3	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.1	10.2	6.9	2.3	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.1	10.3	6.8	2.3	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.9	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	1.2	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.9	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	1.1	0.6	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	8.9	12.2	17.9	50.2	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.6	14.1	10.4	15.4	43.8	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	9	10.6	18.9	50.5	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.1	13.3	10	15	44.9	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	7.8	11.5	17.8	50.8	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.8	13.5	9.7	15.7	44.6	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	10.9	9.1	10.3	18.3	50.7	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	16.9	13.5	10.7	15.2	43	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	3.9	4.7	5.7	11.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	3.8	4.8	5.7	11.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	3.9	4.8	5.8	11.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	3.8	4.7	5.7	11.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.9	4.7	5.7	11.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.8	4.8	5.7	11.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.8	3.9	4.7	5.7	11.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	3.9	4.8	5.7	11.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.7	1.4	2.3	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.7	1.4	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.4	2.4	1.4	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.3	50.9	14.6	1.7	0	8.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.9	50.1	13.2	1.2	0	8.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.7	49.3	14.9	1.5	0	8.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	27	50.5	13.1	1	0	8.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	25	49.3	15.9	1.5	0	8.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.4	50.5	12.6	1.1	0	8.4
U2 - Efficacy additive alpha prior Beta(1, 1)	25.3	49.5	15.2	1.6	0	8.4
U3 - Efficacy additive alpha prior Beta(1, 1)	27.5	48.9	13.7	1.5	0	8.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	10.4	6.8	2.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	10.5	6.8	2.2	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	10.4	6.9	2.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	10.3	6.8	2.3	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.1	10.4	6.9	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	10.4	6.8	2.2	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.2	10.5	6.8	2.2	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.1	10.4	6.9	2.2	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1.2	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1.2	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1.2	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1.2	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.2	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.2	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.2	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.2	0.7	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.6	45.8	27.8	6.5	1	4.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.5	47	22.3	5.6	0.3	4.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.8	45.8	27.3	6.9	0.9	4.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.4	47.8	23	5.1	0.4	4.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.1	45.2	27.9	7.7	0.7	4.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.2	50	21.2	5.9	0.4	4.3
U2 - Efficacy additive alpha prior Beta(1, 1)	14.6	45.6	27.5	7	1	4.3
U3 - Efficacy additive alpha prior Beta(1, 1)	19.1	46.5	24	5.6	0.5	4.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.8	8.7	8.1	4.6	2.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.9	8.7	8.2	4.6	2.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.8	8.7	8.1	4.6	2.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	8.7	8.2	4.6	2.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	8.8	8.1	4.6	2.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.7	8.7	8.2	4.6	2.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.8	8.8	8.1	4.5	2.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.9	8.7	8.2	4.5	2.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.7	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1.8	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.7	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.5	1.8	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.7	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.8	1.5	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.7	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.7	1.5	0.5	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	58	1.6	0	0	0	40.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	58.1	1.5	0	0	0	40.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	57.7	1.6	0	0	0	40.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	58.6	1.5	0	0	0	39.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	57.7	1.7	0	0	0	40.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.3	1.6	0	0	0	40.1
U2 - Efficacy additive alpha prior Beta(1, 1)	58.8	1.3	0	0	0	39.9
U3 - Efficacy additive alpha prior Beta(1, 1)	58.4	1.4	0	0	0	40.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.5	2.2	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.6	2.2	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.6	2.2	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.6	2.2	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.6	2.2	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.7	2.2	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20.6	2.2	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.6	2.2	0.6	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	0.7	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.6	53.1	12.6	1	0	4.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.8	52.5	9.9	1.1	0	4.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.2	53.7	12.2	1.1	0	4.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.6	52	10	0.6	0.1	4.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.3	52.1	12.3	1.3	0.1	4.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33	51.4	9.6	1.2	0	4.8
U2 - Efficacy additive alpha prior Beta(1, 1)	28	54.5	12.1	0.7	0	4.7
U3 - Efficacy additive alpha prior Beta(1, 1)	32.2	52.3	9.9	1.1	0	4.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	11.9	6	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	12	6.1	1.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	12	6.1	1.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	12	6.1	1.7	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	12	6	1.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.2	12.1	6.1	1.7	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.3	12	6.1	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.2	12	6.1	1.8	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.9	1.2	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.9	1.2	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.9	1.2	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.9	1.2	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.9	1.2	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.9	1.2	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.9	1.1	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.9	1.1	0.5	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21	3.7	0.2	0	0	75.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.3	3.3	0.2	0	0	75.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21	3.9	0.3	0	0	74.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.5	3.4	0.1	0	0	76
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.8	3.6	0.3	0	0	75.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.1	4.2	0.1	0	0	74.6
U2 - Efficacy additive alpha prior Beta(1, 1)	20.9	3.7	0.5	0	0	74.9
U3 - Efficacy additive alpha prior Beta(1, 1)	20.8	4.5	0	0	0	74.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.6	3.1	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.7	3.1	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.6	3.1	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.5	3.1	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	3.2	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	3.1	1	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.6	3.1	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.6	3.1	0.9	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.4	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.4	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.4	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.4	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.4	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.4	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.4	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.4	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	1.1	6.3	46.9	44.4	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.4	2.4	7.6	47.5	40.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	1.5	5.7	46.3	45.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.5	2.6	6.8	48.5	39.6	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	1.8	5.9	46.5	44.7	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.6	2.3	8	47.8	39.3	0
U2 - Efficacy additive alpha prior Beta(1, 1)	1	1.2	5.9	46.1	45.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	2.9	2.4	7.8	46.7	40.2	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.2	3.8	8.5	11.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.2	3.7	8.4	11.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	3.2	3.7	8.4	11.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	3.2	3.7	8.4	11.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.3	3.2	3.7	8.5	11.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.3	3.2	3.7	8.4	11.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.3	3.2	3.8	8.4	11.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.3	3.2	3.7	8.4	11.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.1	3.2	1.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.9	31.6	28	3.8	0	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	41.9	30.5	23.9	3	0	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.9	30.9	27.6	3.9	0	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	41.9	29.5	24.9	3	0	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.3	30.9	28.3	3.8	0	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	41.4	30.1	24.7	3.1	0	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	34.5	32.2	28.6	4	0	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	41	31.5	23.4	3.4	0	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	8.3	8.6	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	8.4	8.7	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	8.5	8.5	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	8.3	8.7	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	8.4	8.5	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	8.3	8.7	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	8.4	8.7	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.5	8.3	8.7	2.9	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.1	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.2	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.1	1.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.1	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.1	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.1	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.1	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.1	1.4	0.8	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.2	27.2	23.4	4.6	0	10.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	36.9	28.3	19.6	4.7	0	10.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.2	27.3	24.3	4.8	0	10.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.7	28.3	20.3	4.1	0	10.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.8	27.8	22.7	5.1	0	10.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.7	27.9	20.4	4.6	0	10.4
U2 - Efficacy additive alpha prior Beta(1, 1)	35.1	27.2	23	4.2	0	10.5
U3 - Efficacy additive alpha prior Beta(1, 1)	36.9	28.8	19.1	4.8	0	10.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	7.3	7.2	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	7.4	7.4	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	7.3	7.3	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	7.3	7.2	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	7.4	7.1	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	7.3	7.3	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.9	7.4	7.2	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	11	7.5	7.2	2.8	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.1	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3	1.1	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3	1.1	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3	1.2	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3	1.2	0.8	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.7	26.1	38.6	6.5	0.7	5.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.5	25.9	34.5	6.1	0.6	5.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.2	26.9	37.3	6.6	0.5	5.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.5	27.7	33.8	6.2	0.4	5.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.7	25.9	38.4	7.1	0.5	5.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.3	25.7	34.6	6.6	0.4	5.4
U2 - Efficacy additive alpha prior Beta(1, 1)	22.7	25.9	38.5	7	0.5	5.4
U3 - Efficacy additive alpha prior Beta(1, 1)	27.4	27	33.7	6.3	0.2	5.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	6.9	9	3.9	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	6.9	9.1	3.8	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	7	9	3.9	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	6.8	9.3	3.9	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.2	6.9	9.1	3.9	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.2	6.9	9.2	3.9	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.2	6.8	9.1	3.9	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.2	6.9	9.1	3.9	1.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.6	1	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.2	35.6	6.1	0	0	33.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.6	35.5	5.5	0.1	0	33.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.8	34.6	5.8	0	0	33.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.2	36	4.7	0.1	0	33
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.9	35.2	5.8	0.1	0	33
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	34.6	5.6	0	0	33.3
U2 - Efficacy additive alpha prior Beta(1, 1)	26.2	35.6	5.2	0.1	0	32.9
U3 - Efficacy additive alpha prior Beta(1, 1)	25.9	36	4.5	0.3	0	33.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	8.6	3.9	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	8.7	3.9	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.4	8.6	3.9	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.4	8.7	4	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.3	8.8	3.9	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.4	8.6	4	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.2	8.9	3.9	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.4	8.6	3.9	0.9	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	0.7	0.3	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.5	24.2	3.4	0.3	0	2.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	72.4	21.1	3.3	0.5	0	2.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.4	25.6	3.1	0.2	0	2.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	71.6	22.7	2.8	0.2	0	2.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	70.5	23.5	3	0.2	0.1	2.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	73.5	20.8	2.7	0.3	0	2.7
U2 - Efficacy additive alpha prior Beta(1, 1)	69.4	24.3	3.3	0.3	0	2.7
U3 - Efficacy additive alpha prior Beta(1, 1)	71.4	22.5	3	0.4	0	2.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18	7.6	3.3	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.2	7.5	3.3	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.1	7.5	3.3	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.2	7.5	3.3	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	7.5	3.3	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	7.4	3.3	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.1	7.5	3.2	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.1	7.5	3.3	0.5	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2	0.7	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.1	17.9	17.5	34.5	10.2	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.3	21.4	13.4	28.5	7.6	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.3	18.8	17.3	32.8	11	0.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.7	23	13.9	27.7	7.9	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.6	17.9	17	33.9	9.8	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.2	22.3	13.5	28.2	8	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	20.8	18.1	17.3	33	10	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	26.9	22.7	13.8	28.2	7.6	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	4.8	6	8	6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	4.7	6	8.1	5.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	4.8	6	8.2	5.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	4.7	6	8.2	5.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	4.7	6	8.2	6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	4.7	6	8.1	6	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.1	4.7	6.1	8.1	5.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.1	4.8	6	8.1	5.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	1.6	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.6	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.9	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	1.6	2.3	0.9	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	31.5	28.3	25.3	7.6	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	37.5	28.7	20	4.8	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.8	31.4	29	25.5	7.2	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.4	38.9	27.8	20.3	4.5	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.1	31.5	28.4	25.4	7.5	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.1	38.7	27.5	20.2	4.4	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	6.8	31.4	29.2	24.6	7.9	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	9.3	37.7	28.2	20.6	4.1	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.6	6.5	8	5.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.6	6.5	7.9	5.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	5.5	6.6	8	5.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	5.7	6.5	7.9	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	5.6	6.5	7.9	5.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	5.6	6.4	8	5.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.1	5.6	6.5	7.9	5.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.1	5.6	6.5	8	5.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.3	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.3	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.3	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.4	1.9	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	15.7	32.4	29.2	11.5	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	19.7	31.7	24.5	7.1	0.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	15.8	32.3	29.8	11.4	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.8	21.2	29.8	24.4	7.2	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	15.7	33.5	28.2	11.5	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	21.4	30.8	24.1	7.2	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	10.6	14.5	32.9	28.6	12.8	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	15.8	21.4	30.8	24.8	6.6	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	4.3	7.5	7.4	6.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	4.3	7.6	7.4	6.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	4.3	7.4	7.5	6.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	4.3	7.4	7.5	6.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.3	7.4	7.5	6.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.3	7.3	7.5	6.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	4.3	7.4	7.5	6.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	4	4.3	7.4	7.5	6.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.9	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.9	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.9	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.9	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.9	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	1.9	2.7	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.9	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.9	2.7	1.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	6.1	14	40.7	29.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.3	8.5	13.7	38.2	25.3	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	5.9	13.3	40.5	29.9	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15	8.5	13.5	37.6	25.4	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	6.6	12.9	40.4	29.9	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	7.6	13.4	39.6	26.1	0
U2 - Efficacy additive alpha prior Beta(1, 1)	10.1	5.6	14.1	40.3	29.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	14.7	8.8	13.6	37.6	25.3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	3.6	5.2	8.3	9.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.5	5.2	8.3	9.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	3.6	5.2	8.3	9.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.5	5.2	8.3	9.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	3.5	5.2	8.2	9.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.6	5.2	8.3	9.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.6	3.6	5.2	8.3	9.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.6	5.2	8.3	9.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.8	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.7	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.8	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.8	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1.4	2.7	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1.4	2.8	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.3	15.7	44.3	16.4	4.5	1.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.4	16.2	40.4	15.4	2.7	1.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	16.1	44.5	17.3	4.3	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.9	17.1	38.8	15.5	2.9	1.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.8	15.7	43.8	15.7	4.2	1.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.1	16.5	40.3	14.9	3.4	1.8
U2 - Efficacy additive alpha prior Beta(1, 1)	17.1	17.1	43.3	16.9	3.8	1.8
U3 - Efficacy additive alpha prior Beta(1, 1)	24.5	17.4	39.3	14.2	2.7	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	6.1	9.4	5.5	3.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	6	9.4	5.6	3.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	6.1	9.3	5.6	3.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	6.1	9.3	5.6	3.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	6.1	9.4	5.4	3.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	6.1	9.4	5.5	3.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.5	6.1	9.3	5.6	3.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.5	6	9.3	5.6	3.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.2	1.9	1.6	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.2	1.9	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.2	1.9	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.3	1.9	1.6	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.2	1.9	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.3	1.9	1.6	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.2	1.8	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.3	1.8	1.6	0.6	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.1	40.3	33.8	11.2	2.5	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.3	44.6	29.4	8.9	1.7	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12	39.8	34.6	10.7	2.8	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.1	45.4	28.2	8.5	1.7	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	39.2	34.8	11.3	2.7	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.1	43.7	29.6	9.6	1.9	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	11.4	41.8	33.8	10.8	2.1	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	15.2	44.5	29.3	9.1	1.8	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	8.3	9	5.3	2.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	8.2	9	5.3	2.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	8.2	9.1	5.2	2.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	8.3	9	5.3	2.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	8.3	8.9	5.3	2.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	8.3	8.9	5.3	2.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	8.4	9	5.2	2.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.6	8.4	8.9	5.3	2.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.1	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.6	1.4	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.2	40	12.9	5.7	2.1	15.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.7	40.1	11.1	4.9	1.3	14.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.1	39.7	13.1	6	2.2	14.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.8	39	10.5	4.7	1.8	15.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.5	39.7	13.5	5.9	2.3	15.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29	38.6	11.1	4.8	1.7	14.8
U2 - Efficacy additive alpha prior Beta(1, 1)	23.2	39.7	15	4.9	2.3	14.9
U3 - Efficacy additive alpha prior Beta(1, 1)	28.7	39.3	10.9	4.7	1.3	15.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	8.6	5.4	3	1.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	8.7	5.3	3	1.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	8.5	5.4	3	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	8.4	5.5	3	1.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	8.6	5.5	2.9	1.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	8.6	5.5	2.8	1.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.7	8.4	5.5	2.9	1.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.6	8.5	5.5	2.9	1.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	1.1	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.5	1.1	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.5	1.1	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.5	1.1	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.5	1.1	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.5	1.1	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	1.1	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.5	1.1	0.9	0.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.6	46.6	12.9	2.5	0.5	6.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.6	46.1	9.7	2.3	0.3	7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.2	47.4	13.8	2.2	0.6	6.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	37	42.9	11.2	1.7	0.3	6.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.6	47.4	14.3	2.4	0.6	6.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.5	45.4	11.1	1.9	0.2	6.9
U2 - Efficacy additive alpha prior Beta(1, 1)	29.5	47	13.2	2.5	0.6	7.2
U3 - Efficacy additive alpha prior Beta(1, 1)	35.6	44.7	10.6	1.6	0.4	7.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	10.2	6.4	2.5	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	10.2	6.4	2.5	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	10.2	6.4	2.5	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	10.1	6.4	2.5	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	10.3	6.3	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	10.3	6.3	2.4	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.8	10.2	6.3	2.5	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.8	10.2	6.3	2.5	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	1.3	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4	1.2	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.9	1.3	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.9	1.3	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.9	1.2	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.9	1.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1.3	0.9	0.2	